

Unveiling the Passion: A Journey into Computer Science Education

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In the rapidly advancing field of technological advancement, educators serve as the guiding lights, shaping the minds of tomorrow's innovators. Through an insightful interview with Ms. Novillo, a distinguished software developer turned fervent computer science educator at Stuyvesant High School, we examine of her transition from the realm of coding to the realm of teaching, exploring her motivations, aspirations, and the profound impact she seeks to impart upon her students.

Getting Started

Ms. Novillo's trajectory into the realm of computer science traces back to her formative years in high school. Under the mentorship of an inspiring instructor, she discovered her passion for coding and embarked on a journey of exploration and discovery as she learned topics such as software development. College presented new challenges, however.

"What my mom always wanted was for me to become a doctor," Novillo said. "But I said no, that is not for me."

Ms. Novillo opted instead for what she considers a dynamic field: computer science. The initial spark lit by her instructor kindled a lasting interest in the field, prompting her to defy familial expectations and pursue a career path aligned with her passions.

With a robust background spanning fourteen years in software development, Ms. Novillo embarked on a new chapter in her professional journey as an educator, a dream she had nurtured since college, where she often helped peers overcome challenges. During her time in university, she realized she not only enjoyed problem-solving but also guiding others to see solutions more clearly. 14 years later, Ms. Novillo found herself at Stuyvesant High School, where she could impart knowledge and foster intellectual curiosity among her students. Inspired by the instructors who had guided her, she assisted in teaching at Stuyvesant and soon took on the full role of a teacher. Offering courses ranging from AP Computer Science to software development, Ms. Novillo endeavors to equip her pupils with not only technical proficiency but also essential skills such as creativity, critical thinking, and problem-solving. She accomplishes this through an interactive lesson-based

class. Usually starting with a presentation about the day's topic, she helps students gain an understanding of the material by allowing them to talk in pairs and use whiteboards to display trace diagrams. The remaining portion of the period, ranging from half to three-quarters of the period, is spent working on code for labs and classwork.

Looking Back

Reflecting on her inaugural year at Stuyvesant, Ms. Novillo exudes enthusiasm and gratitude for the enriching experience she has encountered. Immersed in a supportive environment with dedicated colleagues and eager learners, she perceives every challenge as an opportunity for personal and professional growth. Despite the constraints posed by time, her unwavering commitment to inspiring and empowering her students remains steadfast.

When contemplating the rewards and challenges of teaching, Ms. Novillo's passion for her vocation becomes palpable. She derives fulfillment from witnessing the intellectual growth and achievements of her students, cherishing each moment spent within the classroom. However, she acknowledges the demanding nature of her profession, where time constraints present obstacles yet serve as catalysts for continual improvement and innovation. Between the commute to Stuyvesant and the grading of labs and tests in late hours, the job can limit one's ability to spend time on their own hobbies. On the other hand, that time is well spent, contributing to the education of students as they begin their journeys in CS.

Ms. Novillo places a strong emphasis on the value of research in the field of computer science education. Recognizing that not everything can be taught within the confines of a single class period, she encourages her students to delve deeper into topics of interest, fostering a culture of curiosity and exploration. With constant reminders to take the time to do additional research on course material and search for answers themselves before asking others, Ms. Novillo tries to promote academic growth.

"Your teacher can give you their basic idea," she said. "But you can expand that knowledge by researching because this is a great field." Through diligent research and self-directed learning, students can augment their understanding of complex concepts and stay abreast of advancements in the ever-evolving landscape of technology.

Alternate Paths

In a personal insight, Ms. Novillo shares her affinity for mathematics, particularly calculus, highlighting the breadth of her academic interests. Despite her steadfast dedication to computer science, she acknowledges the allure of mathematics and envisions a parallel universe where she imparts the intricacies of calculus to eager minds.

"When I decided to become a teacher, one of my first choices was being a Spanish teacher because that is my native language, or becoming a math teacher."

As she remained on the STEM path, Ms. Novillo's decision aids many students today.

Ultimately, Ms. Novillo extends a heartfelt invitation to students, beckoning them to embark on a transformative journey through the realms of computer science education.

In her classroom, learning goes beyond traditional teaching methods, evolving into a shared journey driven by curiosity, creativity, and the excitement of discovering new things. Ms. Novillo's words resonate, echoing the sentiment that computer science, as a beacon of hope, illuminates the path to a brighter, more empowered future.